

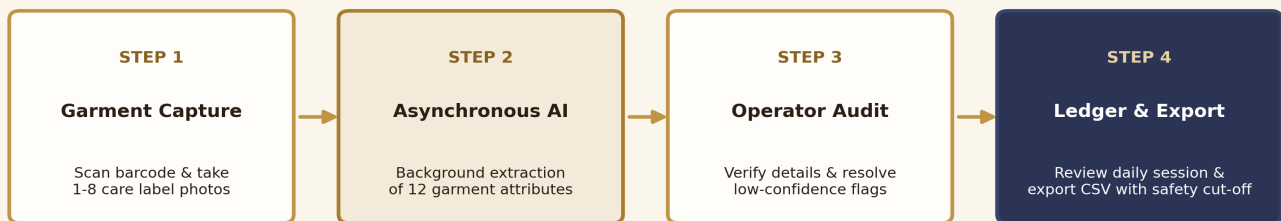
LABEL READER SYSTEM

OPERATOR USER MANUAL
& SYSTEM INTAKE GUIDE

System Overview & Production Architecture

The **Label Reader Intake System** (Apparel Vision) is a high-performance production utility designed for high-throughput garment cataloging, barcode association, optical care-tag extraction, and daily audit ledger management. The intake application connects seamlessly with an advanced middleware Vision AI backend. This AI processes images asynchronously in the background so that the operator is never blocked and can continue scanning subsequent garments immediately.

The Four-Step Production Pipeline



Step 1: Intake / Capture: Operators scan a barcode, capture up to 8 high-resolution photos of garment labels, and select a primary 'Key Photo' before committing the item.

Step 2: Asynchronous AI Extraction: Scans are batched and processed in the background by the vision backend without interrupting manual floor operations.

Step 3: Verification Workspace: Operators review the 12 extracted attributes, verify details, resolve low-confidence warning flags, and commit records to the daily ledger.

Step 4: Daily Ledger & Export: Operators audit the active session, duplicate multi-packs, and complete the certified two-step CSV export and cut-off handshake.

1. Garment Capture & Intake Workflow

Step 1.1: Scan or Enter Garment Barcode

Open the **Capture** workspace from the bottom navigation bar. Tap the Garment Barcode / ID input field. To scan a barcode, tap the **Scan** button to open the camera barcode reticle. Point your viewfinder at the barcode/QR code until the green frame confirms recognition. The barcode will automatically populate into the field. If needed, barcodes can also be typed manually using the keyboard.

Step 1.2: Photographing Care Labels (1 to 8 Photos)

Position the wash tag, brand neck label, material composition tag, or price tag inside the camera viewfinder. **Tap anywhere on the viewfinder screen** to set the camera auto-focus on tiny care label text. If lighting is dim, tap the **Torch** (flash) icon in the header bar. Press the large Shutter Button (or tap Gallery Import to select local files). You can capture up to 8 photos for different angles or multi-page care labels (e.g., brand tags, wash symbols, RN numbers).

Designate Key Photo: In the bottom photo filmstrip, tap the **Star (★)** icon on the thumbnail that best represents the garment. This photo serves as the primary catalog thumbnail and CSV reference image. To delete a blurry photo, tap the **Close (X)** icon on its thumbnail.

Step 1.3: Committing the Item ('Finish Item')

All captured photos remain ephemeral in device memory until committed. Once you have entered a barcode and captured at least one care label photo, tap the **Finish Item** button. This saves the item to the local SQLite database and immediately queues it for background AI vision processing. The capture screen automatically resets to a clean state so you can immediately process the next garment.

BEST PRACTICES FOR HIGH-QUALITY CAPTURE

- **Keep it Flat:** Ensure tags and care labels are flat, wrinkle-free, and well-lit to maximize AI extraction accuracy.
- **Verify Auto-Focus:** Tap the camera view to focus on small text before snapping. Blurry images cause AI failures.
- **Cover All Details:** For garments with multi-paged wash tags, snap separate photos of each page. Ensure you always mark the main front brand tag with the **Star (★)** as the Key Image.

2. Verification Workspace & Field Audit

Step 2.1: Accessing Completed Extractions

Select the **Review** tab in the bottom navigation bar. Under the **Ready to Review** list, you will see all garment cards where the Vision AI extraction is complete. Tap on any card to open the full-screen **Care Label & Composition Audit Modal**.

Step 2.2: Auditing the 12 Extracted Attributes

The modal features an interactive image inspector on top (with thumbnails of all label photos) and 12 editable text fields. Operators must review and verify the following 12 attributes against the tag images:

1. Brand Name (e.g. Zara, Levi's, Nike)	7. Color (e.g. Navy Blue, Heather Grey)
2. Category (e.g. Tops, Bottoms, Knitwear)	8. Material Composition (e.g. 100% Cotton)
3. Sub-Category (e.g. Crewneck Tee, Slim Denim)	9. Country of Origin (e.g. Made in Vietnam)
4. Gender (Men, Women, Unisex, Kids)	10. Original Price (e.g. €79.90, \$120.00)
5. Season (e.g. SS24, FW24, All-Season)	11. Netto Weight (in kilograms, e.g. 0.32 kg)
6. Size (e.g. M, 32/34, 40R, UK 10)	12. Brutto Weight (in kilograms, e.g. 0.35 kg)

Step 2.3: Understanding Visual Confidence Flags

To maximize speed, the application flags fields with different background colors based on AI confidence levels:

- **Normal Gold / Linen Fields:** High extraction confidence (> 70%). These fields are highly likely to be correct. Quickly glance to confirm.
- **Rust / Orange Highlighted Fields:** Low extraction confidence (< 70%) or the field was completely undetected. The operator is **required** to cross-reference the label image, verify the value, and type in the correct attribute manually.

Once all fields are verified, tap **Confirm & Save to Ledger**. This promotes the item into the active Daily Ledger.

MANDATORY CONFIDENCE CHECKPOINT

Fields marked in **Rust** require immediate attention and represent a manual validation checkpoint. Never tap 'Confirm & Save' without looking at the care label image to verify all low-confidence fields. Ensuring data integrity at this stage prevents corrupted CSV reports and catalog errors.

3. Daily Ledger & Safe CSV Export

Step 3.1: Reviewing Active Records & Cloning

The **Daily Ledger** tab displays all garments verified during the current operational shift. For multi-packs or identical garments that differ only by barcode, operators can use the **Cloning** feature: Tap the **Copy Icon** on the garment card, type or scan the new barcode, and tap **Duplicate**. All 12 attributes are instantly copied to the new item, bypassing manual capture and AI wait times entirely.

Step 3.2: Safe Two-Step CSV Export Handshake

To prevent duplicate exports or inadvertent data loss, the system employs a rigid two-step cut-off process:

- **Step A: Initiate Export:** Tap **Export CSV** in the top header. Choose your export destination (e.g. Drive, Email, Local Downloads) in the Android native Share sheet. Return to the app after the transfer.
- **Step B: Safe Cut-Off Dialog:** A prompt will ask: *'Confirm CSV Delivery & Cut-Off: Have all exported items been successfully received or saved?'*
 - Tap **Confirm Cut-Off** to advance the shift session. This stamps all exported items with a batch ID and timestamp, and moves them to the **All History Archive** tab, clearing your active workspace.
 - Tap **Keep in Active Session** to cancel the cut-off if you need to re-export or add more items.

4. Helpdesk & Troubleshooting Reference

Operator Reported Issue	Likely Root Cause	Step-by-Step Operator Solution
' Server Connection Offline ' banner displayed	App cannot reach configured base URL (WiFi down, incorrect network IP).	1. Go to Settings tab. 2. Verify Server Base URL. 3. Tap Demo Simulation Mode if offline fallback is required. Scans remain safe in SQLite.
Camera shows black screen / permission prompt	Operating System denied Camera hardware access permissions.	1. Tap Grant Camera Permission directly on the viewfinder. 2. Or open Android Settings > App Info > Permissions and enable Camera.
Device storage low warning	Accumulated scratch photos from canceled/abandoned sessions.	1. Go to Settings tab. 2. Locate Storage & Unused Photos. 3. Tap Delete Unused Photos Cache to purge scratch files.
Operator accidentally exported CSV twice	Ledger session was not cut off after the first CSV export.	Complete the two-step verification. In the confirmation dialog, tapping Confirm Cut-Off will cleanly advance the session so duplicates cannot occur.
Hard reset / device decommissioning required	Terminal decommissioning or corrupt local testing database state.	1. Go to Settings > DANGER ZONE. 2. Check <i>'I understand all data will be permanently destroyed'</i>. 3. Tap Delete All Data & Photos.